

Akash Karthik

San Jose, CA - akashkarthik473@gmail.com - linkedin.com/in/akash-karthik473 - github.com/akashkarthik473 - 9254069507

EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C/C++, SQL, Java

Backend & Web: FastAPI, Flask, Node.js/Express, React, REST/JSON, WebSockets, real-time streaming, input validation

IoT & Embedded: CAN/DBC signal decoding, python-can, STM32H7, TTC 60 VCU, PX4, real-time firmware, vehicle telemetry

Data & DevOps: SQLite, SQL Server, pandas, NumPy, schema design, Docker, pytest, Jest, GitHub Actions, Git

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead software planning for Python telemetry tooling, controller validation, CAN diagnostics, and shared testing workflows
- Standardize project requirements, test evidence, and technical reviews across dashboard, simulation, and vehicle software projects

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated flight-control software in C on PX4, using structured telemetry logs to reproduce issues and evaluate changes
- Integrated visual-inertial odometry and iterated configuration through repeatable tests and documented results

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Built CAN diagnostics firmware publishing live power and torque frames, status bits, and error counters for real-time bus monitoring
- Developed Python telemetry pipelines to parse vehicle logs, calculate controller response metrics, and generate plots for tuning and validation

Software Development Intern

Sep 2024 - Oct 2024

Zip Intelligence | Tracy, CA

- Implemented frontend validation and clearer error states, then paired with engineers to harden REST endpoints for status codes and edge cases
- Added SQL Server update scripts and structured logs to speed reproduction and issue triage
- Took ownership of a small feature from ticket through verified release, shipping via pull requests with setup and change documentation

PROJECT EXPERIENCE

SR-Wireless-CAN — Wireless Vehicle Telemetry Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, FastAPI, WebSockets, python-can, React, SQLite, Docker Compose

- Led backend development for a wireless vehicle telemetry platform, building a FastAPI and WebSocket service that streams live CAN data to a React dashboard with SQLite persistence
- Proposed and implemented wireless firmware flashing for the vehicle control unit, decoding the undocumented flashing sequence from CAN trace captures and reimplementing it in Python
- Implemented structured DBC upload and merge with automatic conflict resolution, added recorded-trace replay, and containerized the stack with Docker Compose

Tickr — AI-Coached Paper-Trading Game

2025

TypeScript/JavaScript, React, LLM integration, responsive UI

- Built registration, sign-in, game, and history flows for a scenario-driven trading game with an AI coach
- Implemented quest and prompt logic, client-side state management, and a mobile-responsive interface